

1) Random forest Algorithm

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(Not good for small datasets, overfitting)

- Random forest is a non-linear machine learning algorithm used for classification (like "cat or dog") and regression (predicting numbers, like salary).
- The name "forest" comes because it has many trees - and those trees are Decision Trees.

How it works (Simple logic)

1. A Decision Tree is like a series of questions that help to make a decision.
- Eg. - "Is the employee experience > 5 years?"
 - "Is salary > 60k?"

→ It keeps splitting until it reaches an answer.

2. But one decision tree can make mistakes or overfit (learn too perfectly).

3. So instead of one tree, Random forest makes many trees from random parts of the data.

- Eg. - If you have 100 students, one tree might use 30 random students, another uses 40 random students, etc.
- Each tree learns something slightly different.

4. When a new question (new data) comes:-

- Every tree gives its own opinion (its own answer).
 - The forest takes a majority vote (most common answer wins).
- That's why it's called Random forest!

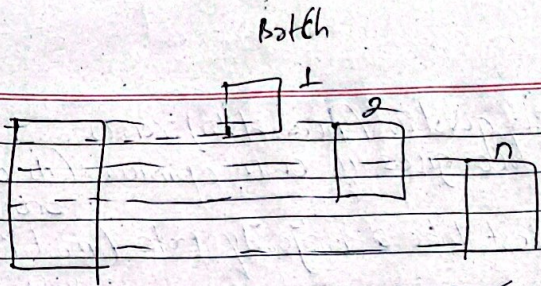
Real-life Example:- (from the video)

Imagine you want to buy a gadget:-

- friend 1 says: "Buy it! It saves money."
- friend 2 says: "Don't buy, it's expensive."
- friend 3 says: "Buy! you can control it from your phone."

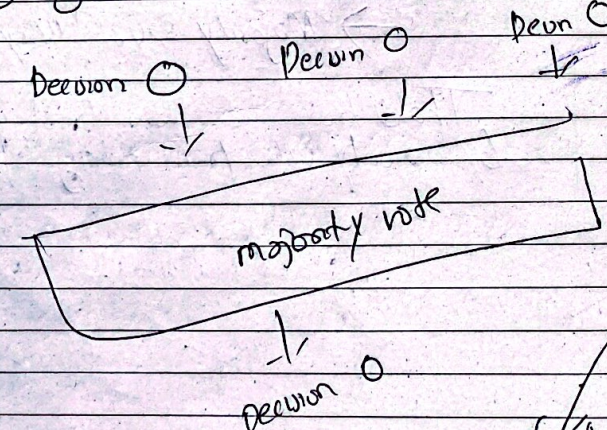
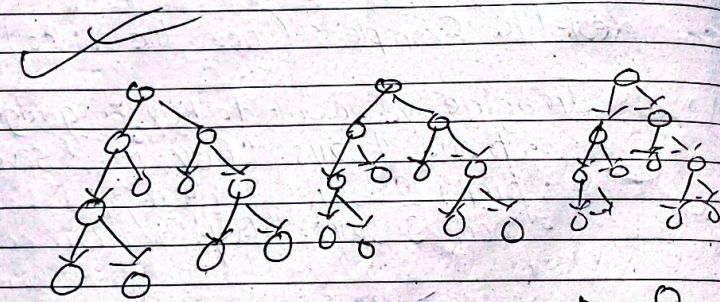
→ Majority says "Buy it", so you do!

That's how Random forest makes decisions - by majority vote from many "friends" (trees).



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from sklearn.metrics
accuracy_score
for i in range(n)
plt.imshow(X_test[i])
(check input)



So instead of
trusting one decision
tree we trust a
group of trees
- that's why
random forest

Code

1. Load dataset

- Use the digits dataset of sklearn (pictures of handwritten numbers 0-9)
- Each image is 8×8 pixels, so each sample has 64 numbers (one for each pixel)

2. Create Dataframe

- Each row = one image (as 64 numbers)
- Add a "target" column = the real digit (truth)

3. Split into training & testing sets

- 80% → training (to teach the model)
- 20% → testing (to check how well it learned)

4. Import RandomForestClassifier

from sklearn.ensemble import RandomForestClassifier

"ensemble" means group of models working together to give a better result than one alone.

5. Train the model

model.fit(X_train, y_train) ~ default n_estimators = 100

6. Check accuracy

model.score(X_test, y_test)

no. of test you make

Cross validation
score
(K-fold)

clustering

K-Means

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from sklearn.datasets import digits

digits = load_digits()
dir(digits)

- More trees (20/30/50) $\rightarrow$ usually better accuracy (but too many may slow it down)!

Confusion Matrix (To see mistakes)

- It compares the true answers vs predicted answers.
- Example: -
 - Model said "8" but truth was "1" $\rightarrow$ that's an error
 - If model said "8" and truth was "8" $\rightarrow$ that's correct
- helps us see which numbers the model finds difficult.

from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_predicted)

cm

truth

predicted value

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,7))

sns.heatmap(cm, annot=True)

plt.xlabel('Predicted') plt.ylabel('Truth')

When to use Random Forest

Use random forest when:-

1. You have lots of data with many features (columns).
Example - predicting salary from experience, age, education, city, etc.
2. You want high accuracy & don't want to worry too much about overfitting (because it averages many trees).
3. You don't know which algorithm is best -
Random forest is a great "safe first choice" because it usually works well on many kinds of data.
4. You want both classification & regression.
 - classification = predict categories (Yes/No, Cat/Dog, Digit 0-9)
 - Regression = predict numbers (Price, Temperature, Salary)
5. Your data is noisy or has missing values.
 $\rightarrow$ Random forest still performs quite well even with messy data.

y_predicted = model.predict(X_test)

Example situations:

Situation	What Random forest does
predict if a person will get a loan	Classification (Yes or No)
predict someone's salary	Regression (Numbers output)
Spam or Not Spam emails	Classification
Recognize handwritten numbers (like in the video)	Classification
Identify flower species (Iris dataset)	Classification

When Not to use Random forest

Use it carefully when:-

1. Speed matters a lot: - It can be slow for very large data because it builds many trees
2. You need very explainable results: - it's harder to understand why it gave a certain answer

compared to a simple decision tree

(Use Random forest when you want accurate & reliable answers - even if your data is a bit messy! ")